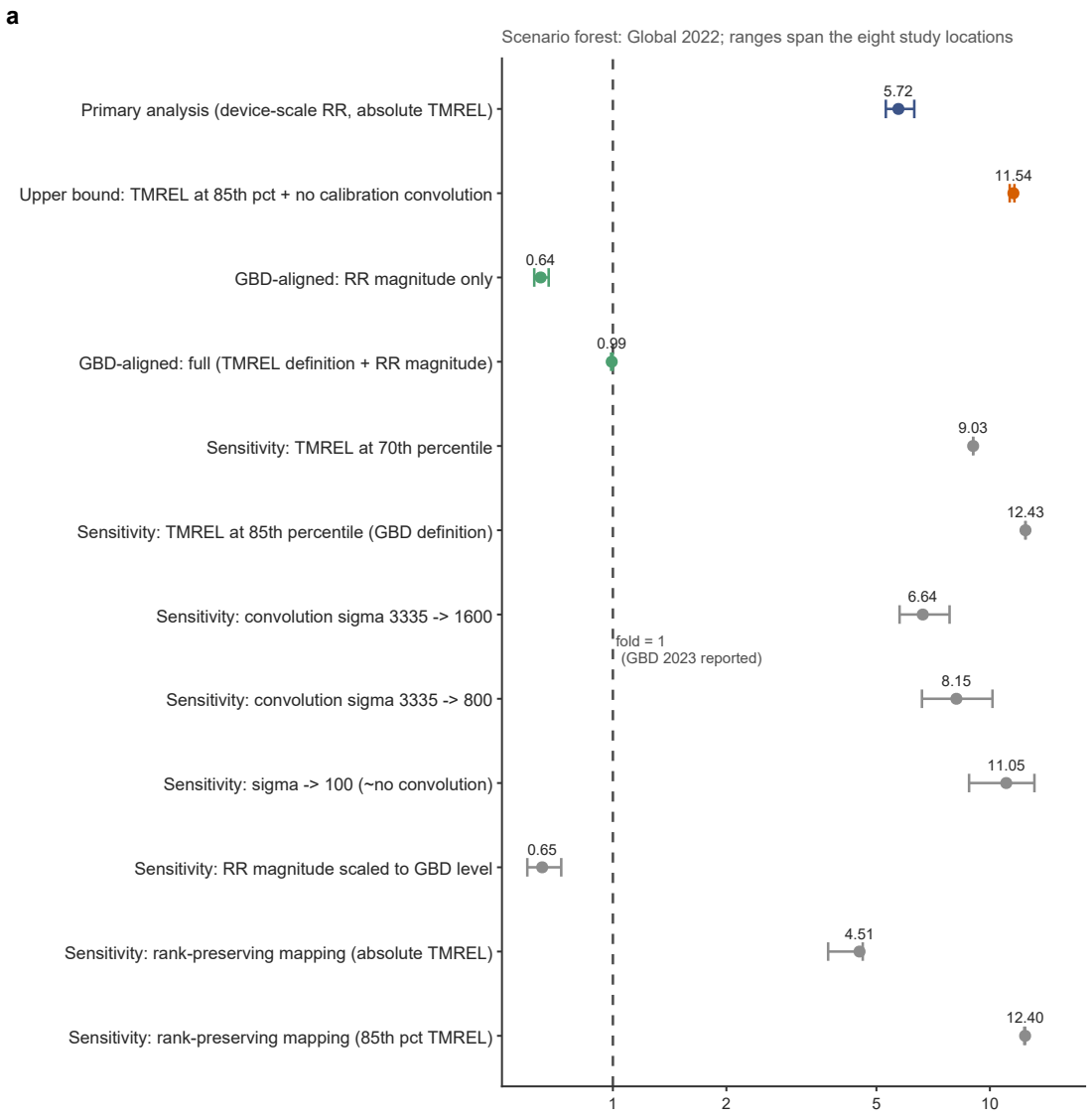
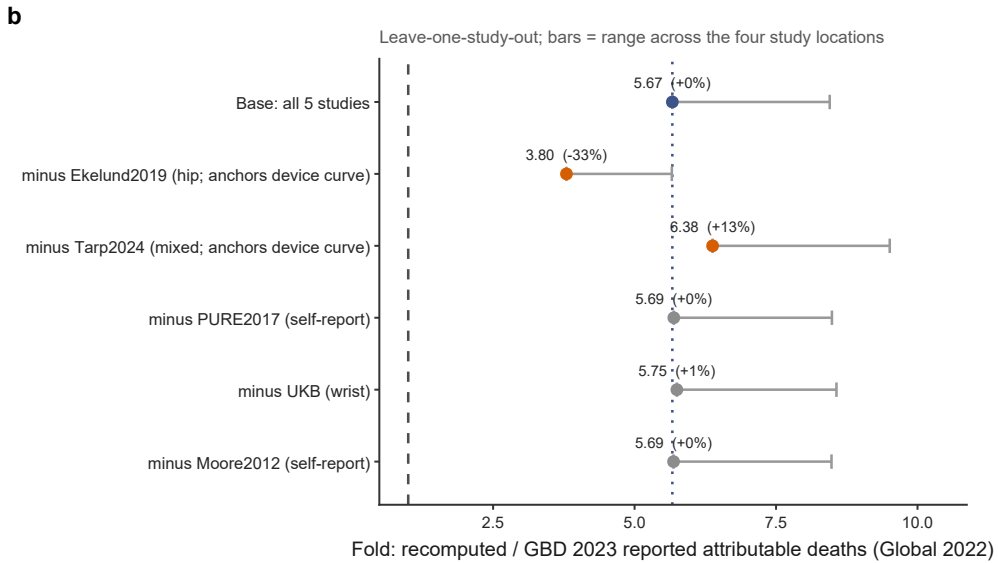
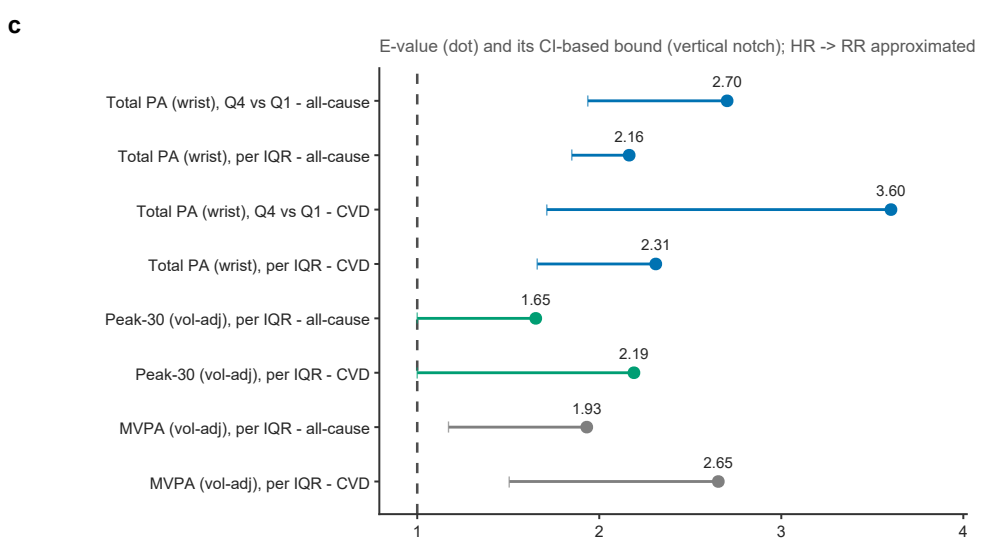

a
Scenario forest: Global 2022; ranges span the eight study locations
Primary analysis (device-scale RR, absolute TMREL)
5.72
Upper bound: TMREL at 85th pct + no calibration convolution
11.54
GBD-aligned: RR magnitude only
0.64
GBD-aligned: full (TMREL definition + RR magnitude)
0.99
Sensitivity: TMREL at 70th percentile
9.03
Sensitivity: TMREL at 85th percentile (GBD definition)
12.43
Sensitivity: convolution sigma 3335 -> 1600
6.64
fold = 1
(GBD 2023 reported)
Sensitivity: convolution sigma 3335 -> 800
8.15
Sensitivity: sigma -> 100 (~no convolution)
11.05
Sensitivity: RR magnitude scaled to GBD level
0.65
Sensitivity: rank-preserving mapping (absolute TMREL)
4.51
Sensitivity: rank-preserving mapping (85th pct TMREL)
12.40
1
2
5
10
Fold: recomputed / GBD 2023 reported attributable deaths (log scale)


b
Leave-one-study-out; bars = range across the four study locations
Base: all 5 studies
5.67 (+0%)
minus Ekelund2019 (hip; anchors device curve)
3.80 (-33%)
minus Tarp2024 (mixed; anchors device curve)
6.38 (+13%)
minus PURE2017 (self-report)
5.69 (+0%)
minus UKB (wrist)
5.75 (+1%)
minus Moore2012 (self-report)
5.69 (+0%)
2.5
5.0
7.5
10.0
Fold: recomputed / GBD 2023 reported attributable deaths (Global 2022)


c
E-value (dot) and its CI-based bound (vertical notch); HR -> RR approximated
Total PA (wrist), Q4 vs Q1 - all-cause
2.70
Total PA (wrist), per IQR - all-cause
2.16
Total PA (wrist), Q4 vs Q1 - CVD
3.60
Total PA (wrist), per IQR - CVD
2.31
Peak-30 (vol-adj), per IQR - all-cause
1.65
Peak-30 (vol-adj), per IQR - CVD
2.19
MVPA (vol-adj), per IQR - all-cause
1.93
MVPA (vol-adj), per IQR - CVD
2.65
1
2
3
4
E-value: minimum confounder association strength to explain away the HR